

Discrete Random Variables

ANSWER KEY



Probability distributions

- 1 A discrete random variable X has the probability distribution: $P(X = 1) = 0.2$, $P(X = 2) = 0.3$, $P(X = 3) = k$, $P(X = 4) = 0.1$. Find k , using the fact that probabilities must sum to 1.
 $0.2 + 0.3 + k + 0.1 = 1$, so $k = 1 - 0.6 = 0.4$.
- 2 A discrete random variable Y has the probability distribution: $P(Y = 0) = 0.1$, $P(Y = 1) = k$, $P(Y = 2) = 2k$, $P(Y = 3) = 0.3$. Find k .
 $0.1 + k + 2k + 0.3 = 1$, so $3k = 0.6$, giving $k = 0.2$.
- 3 A discrete random variable Z has the probability distribution: $P(Z = 1) = 0.25$, $P(Z = 2) = 0.25$, $P(Z = 3) = k$, $P(Z = 4) = 0.15$, $P(Z = 5) = 0.15$. Find k .
 $0.25 + 0.25 + k + 0.15 + 0.15 = 1$, so $k = 1 - 0.8 = 0.2$.

Expected value (mean) of a discrete random variable

- 4 A random variable X has distribution $P(X = 1) = 0.2$, $P(X = 2) = 0.5$, $P(X = 3) = 0.3$. Find $E(X)$.
 $E(X) = \sum xP(X = x) = 1(0.2) + 2(0.5) + 3(0.3) = 0.2 + 1.0 + 0.9 = 2.1$.
- 5 A random variable Y has distribution $P(Y = 0) = 0.1$, $P(Y = 1) = 0.4$, $P(Y = 2) = 0.5$. Find $E(Y)$.
 $E(Y) = \sum yP(Y = y) = 0(0.1) + 1(0.4) + 2(0.5) = 0 + 0.4 + 1.0 = 1.4$.
- 6 A random variable Z has distribution $P(Z = 1) = 0.3$, $P(Z = 2) = 0.3$, $P(Z = 3) = 0.4$. Find $E(Z)$.
 $E(Z) = \sum zP(Z = z) = 1(0.3) + 2(0.3) + 3(0.4) = 0.3 + 0.6 + 1.2 = 2.1$.

Variance of a discrete random variable

- 7 Using the distribution from the previous question ($E(X) = 2.1$), find $\text{Var}(X)$, using $\text{Var}(X) = E(X^2) - [E(X)]^2$.
 $E(X^2) = 1^2(0.2) + 2^2(0.5) + 3^2(0.3) = 0.2 + 2.0 + 2.7 = 4.9$.
 $\text{Var}(X) = 4.9 - (2.1)^2 = 4.9 - 4.41 = 0.49$.
- 8 Using the distribution $P(Y = 0) = 0.1$, $P(Y = 1) = 0.4$, $P(Y = 2) = 0.5$ ($E(Y) = 1.4$), find $\text{Var}(Y)$, using $\text{Var}(Y) = E(Y^2) - [E(Y)]^2$.
 $E(Y^2) = 0^2(0.1) + 1^2(0.4) + 2^2(0.5) = 0 + 0.4 + 2.0 = 2.4$.
 $\text{Var}(Y) = 2.4 - (1.4)^2 = 2.4 - 1.96 = 0.44$.

- 9 Using the distribution $P(Z = 1) = 0.3$, $P(Z = 2) = 0.3$, $P(Z = 3) = 0.4$ ($E(Z) = 2.1$), find $\text{Var}(Z)$, using $\text{Var}(Z) = E(Z^2) - [E(Z)]^2$.
- $$E(Z^2) = 1^2(0.3) + 2^2(0.3) + 3^2(0.4) = 0.3 + 1.2 + 3.6 = 5.1.$$
- $$\text{Var}(Z) = 5.1 - (2.1)^2 = 5.1 - 4.41 = 0.69.$$

Linear transformations of a random variable

- 10 A random variable X has $E(X) = 5$ and $\text{Var}(X) = 3$. Find $E(2X + 4)$ and $\text{Var}(2X + 4)$.
- $$E(2X + 4) = 2E(X) + 4 = 2(5) + 4 = 14. \text{Var}(2X + 4) = 2^2\text{Var}(X) = 4(3) = 12.$$
- 11 A random variable X has $E(X) = 8$ and $\text{Var}(X) = 5$. Find $E(3X - 2)$ and $\text{Var}(3X - 2)$.
- $$E(3X - 2) = 3E(X) - 2 = 3(8) - 2 = 22. \text{Var}(3X - 2) = 3^2\text{Var}(X) = 9(5) = 45.$$
- 12 A random variable Y has $E(Y) = 10$ and $\text{Var}(Y) = 4$. Find $E(5 - 2Y)$ and $\text{Var}(5 - 2Y)$.
- $$E(5 - 2Y) = 5 - 2E(Y) = 5 - 2(10) = -15. \text{Var}(5 - 2Y) = (-2)^2\text{Var}(Y) = 4(4) = 16.$$

The discrete uniform distribution

- 13 A fair six-sided die is rolled, and X is the number shown. Find $E(X)$ and $\text{Var}(X)$, given that for a discrete uniform distribution on $1, \dots, n$: $E(X) = \frac{n+1}{2}$ and $\text{Var}(X) = \frac{n^2-1}{12}$.
- $$n = 6: E(X) = \frac{7}{2} = 3.5. \text{Var}(X) = \frac{35}{12} \approx 2.92.$$
- 14 A fair eight-sided die (labeled 1 to 8) is rolled, and X is the number shown. Find $E(X)$ and $\text{Var}(X)$, using the same formulas.
- $$n = 8: E(X) = \frac{9}{2} = 4.5. \text{Var}(X) = \frac{63}{12} = 5.25.$$
- 15 A fair ten-sided die (labeled 1 to 10) is rolled, and X is the number shown. Find $E(X)$ and $\text{Var}(X)$, using the same formulas.
- $$n = 10: E(X) = \frac{11}{2} = 5.5. \text{Var}(X) = \frac{99}{12} = 8.25.$$

Constructing a probability distribution from a real scenario

- 16 A game involves rolling two fair dice and finding the sum X . Find $P(X = 4)$ and $P(X = 7)$.
- There are 36 equally likely outcomes. $X = 4$: (1, 3), (2, 2), (3, 1), so $P(X = 4) = \frac{3}{36} = \frac{1}{12}$. $X = 7$: (1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1), so $P(X = 7) = \frac{6}{36} = \frac{1}{6}$.
- 17 Using the same game of rolling two fair dice and finding the sum X , find $P(X = 9)$ and $P(X = 12)$.
- $X = 9$: (3, 6), (4, 5), (5, 4), (6, 3), so $P(X = 9) = \frac{4}{36} = \frac{1}{9}$. $X = 12$: (6, 6), so $P(X = 12) = \frac{1}{36}$.

- 18 Using the same game of rolling two fair dice and finding the sum X , find $P(X = 5)$ and $P(X = 6)$.
 $X = 5$: (1, 4), (2, 3), (3, 2), (4, 1), so $P(X = 5) = \frac{4}{36} = \frac{1}{9}$. $X = 6$: (1, 5), (2, 4), (3, 3), (4, 2), (5, 1), so $P(X = 6) = \frac{5}{36}$.

Cumulative distribution function

- 19 A random variable X has $P(X = 1) = 0.15$, $P(X = 2) = 0.35$, $P(X = 3) = 0.25$, $P(X = 4) = 0.25$. Find $P(X \leq 2)$ and $P(X > 2)$.
 $P(X \leq 2) = P(X = 1) + P(X = 2) = 0.15 + 0.35 = 0.5$. $P(X > 2) = 1 - 0.5 = 0.5$.
- 20 A random variable X has $P(X = 1) = 0.1$, $P(X = 2) = 0.2$, $P(X = 3) = 0.3$, $P(X = 4) = 0.4$. Find $P(X \leq 3)$ and $P(X \geq 2)$.
 $P(X \leq 3) = P(X = 1) + P(X = 2) + P(X = 3) = 0.1 + 0.2 + 0.3 = 0.6$.
 $P(X \geq 2) = 1 - P(X = 1) = 1 - 0.1 = 0.9$.
- 21 A random variable X has $P(X = 0) = 0.2$, $P(X = 1) = 0.3$, $P(X = 2) = 0.3$, $P(X = 3) = 0.2$. Find $P(X \leq 1)$ and $P(X \geq 3)$.
 $P(X \leq 1) = P(X = 0) + P(X = 1) = 0.2 + 0.3 = 0.5$. $P(X \geq 3) = P(X = 3) = 0.2$.

A well-designed word problem: a raffle ticket game

- 22 This question has two parts. A raffle sells 200 tickets for 5 dollars each. One ticket wins a prize of 300 dollars, and two tickets each win a prize of 50 dollars. Let X be the net profit (prize won minus the 5-dollar ticket cost) for a single ticket. (a) Write the probability distribution of X . (b) Find $E(X)$, and interpret whether buying a ticket is a fair game.
- (a) $X = 295$ (win big prize) with $P = \frac{1}{200}$; $X = 45$ (win small prize) with $P = \frac{2}{200}$; $X = -5$ (no prize) with $P = \frac{197}{200}$. (b) $E(X) = 295\left(\frac{1}{200}\right) + 45\left(\frac{2}{200}\right) + (-5)\left(\frac{197}{200}\right) = \frac{295 + 90 - 985}{200} = \frac{-600}{200} = -3$. Since $E(X) = -3 \neq 0$, the game is not fair — on average, a player loses 3 dollars per ticket.